

Filesystem Folly | MetaCTF Writeup



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Problem

Previous Submissions

We captured network traffic of someone connecting to a file share on a server and writing to a few files.

The flag is in the flag.png file. Find it!

Hint: see if you can isolate the RPC requests that are responsible for writing files to the file system, and pull out the payload of the one that writes flag.png. Copy Bytes as Raw Binary might be helpful, but there are other valid approaches too.

Filesystem Folly (solved by 385 teams)

200

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This challenge is from the May 2024 Flash CTF. It was first solved 2.1 minutes into the competition by [Myxodile](#). 23.3% of active users solved it during the competition.

Submit!

Please rate this problem: ☆☆☆☆☆

process:

what is RPC?

what is NFS?

Procedures used in NFS service

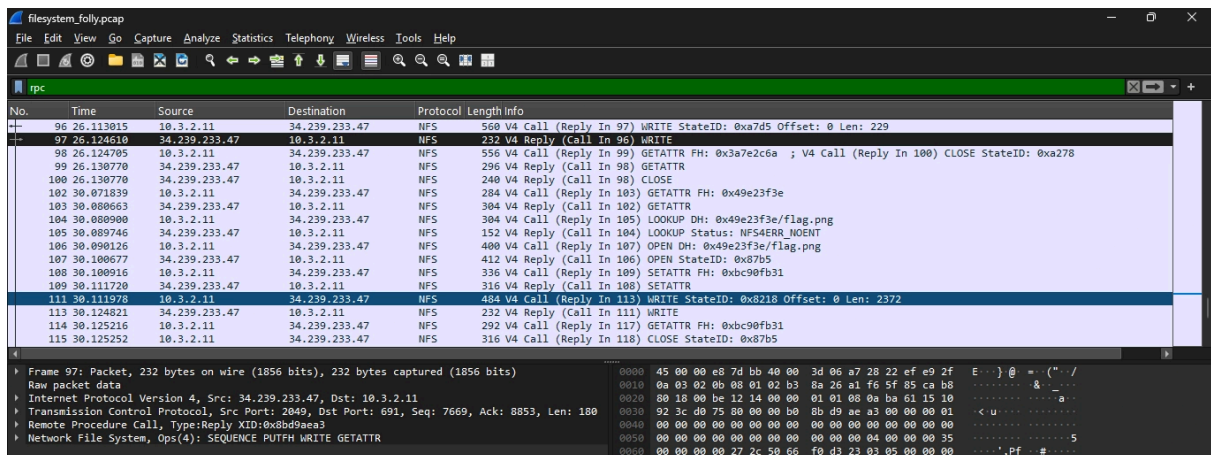
NFS service defines a [list of procedures](#). Here is a list with brief summary of activities. The bottom five RPCs are introduced in v3.

Procedure	Activity
GETATTR(fh)	Returns the attributes of a file, similar to stat syscall.
SETATTR(fh, attr)	Sets the attributes of a file (mode, uid, gid, size, atime, mtime); setting the size to 0 truncates the file
STATFS(fh)	Returns the status of a filesystem, such as block size, number of free blocks. e.g df command.
LOOKUP(dirfh, name)	Returns fhandle and attributes for the named file in the directory specified by dirfh
READ (fh, offset, count)	Reads from a file, with offset and count specified. In v2, the length is up to 8192 bytes; v3 support more.
WRITE (fh, offset, count, data)	Writes to a file, with offset and count specified, as well as a separate field called data. Returns the new attributes of the file after the write.

As mentioned in the challenge, someone connected to a file share on a server and wrote to several files.

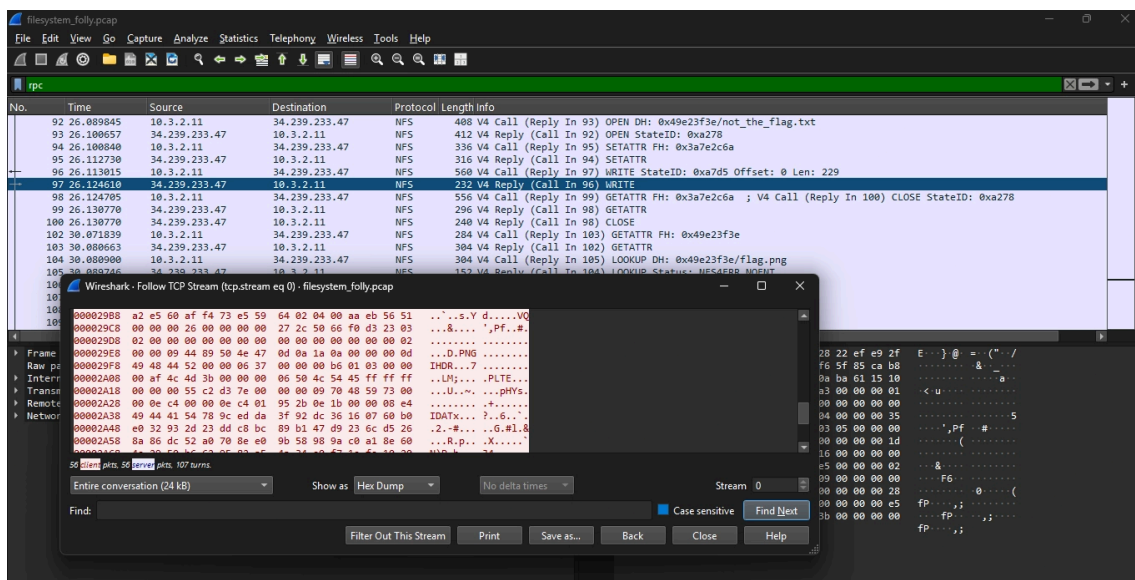
Filtered the RPC / NFS traffic in the provided PCAP for further analysis.

Here, we identified a write operation indicating that this IP address wrote a file to the server.

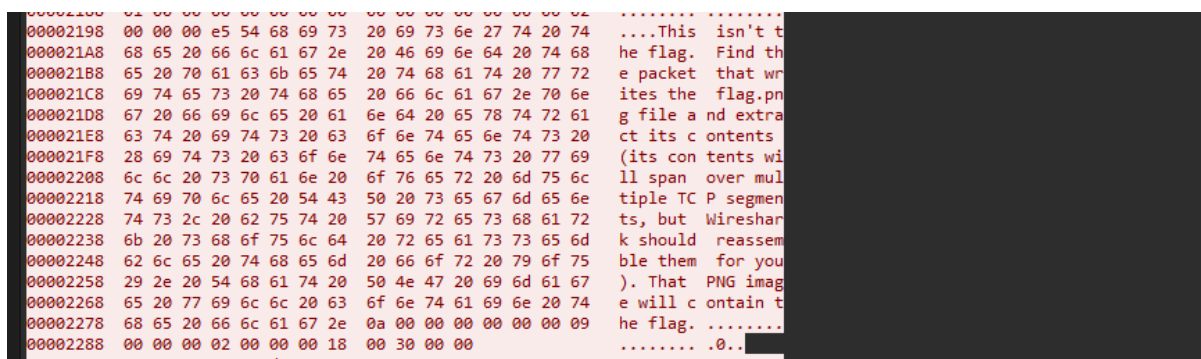


Followed the TCP stream and identified the PNG file's magic bytes.

The file was transferred from the client (10.3.2.11) to the server (34.239.233.47).

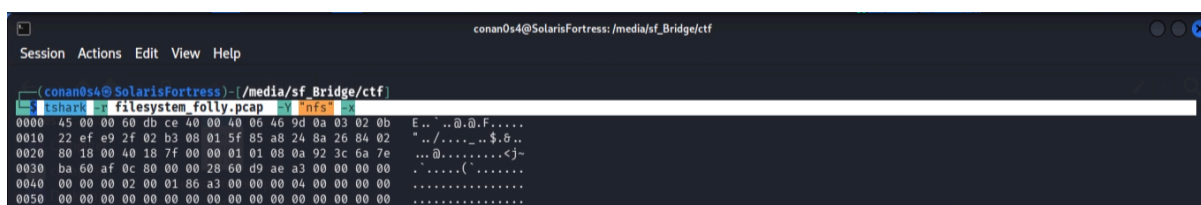


another hint was

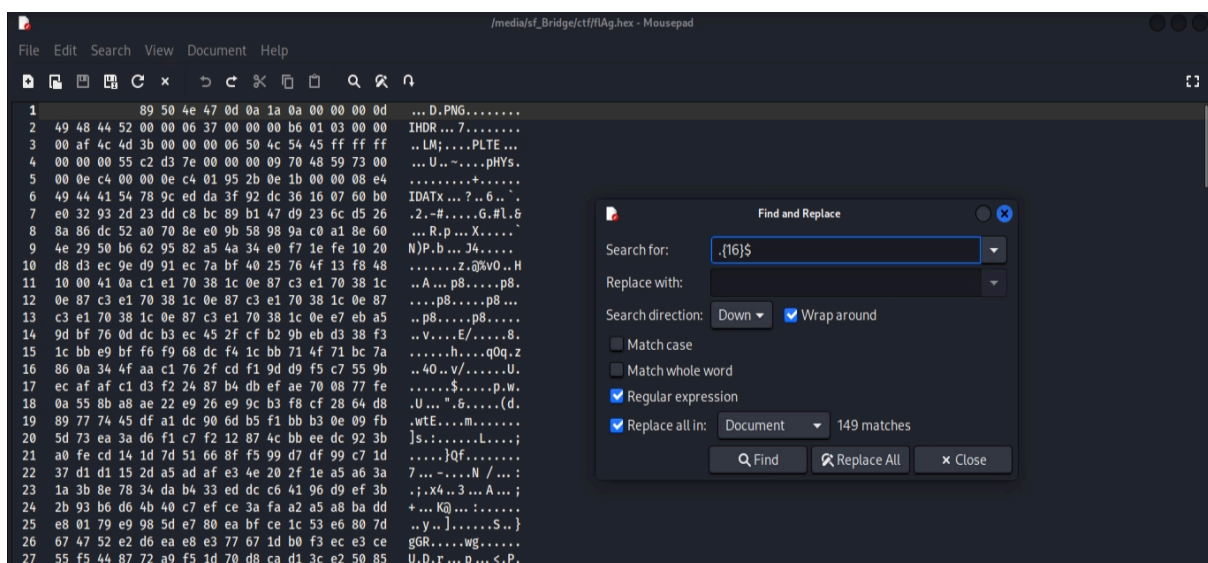
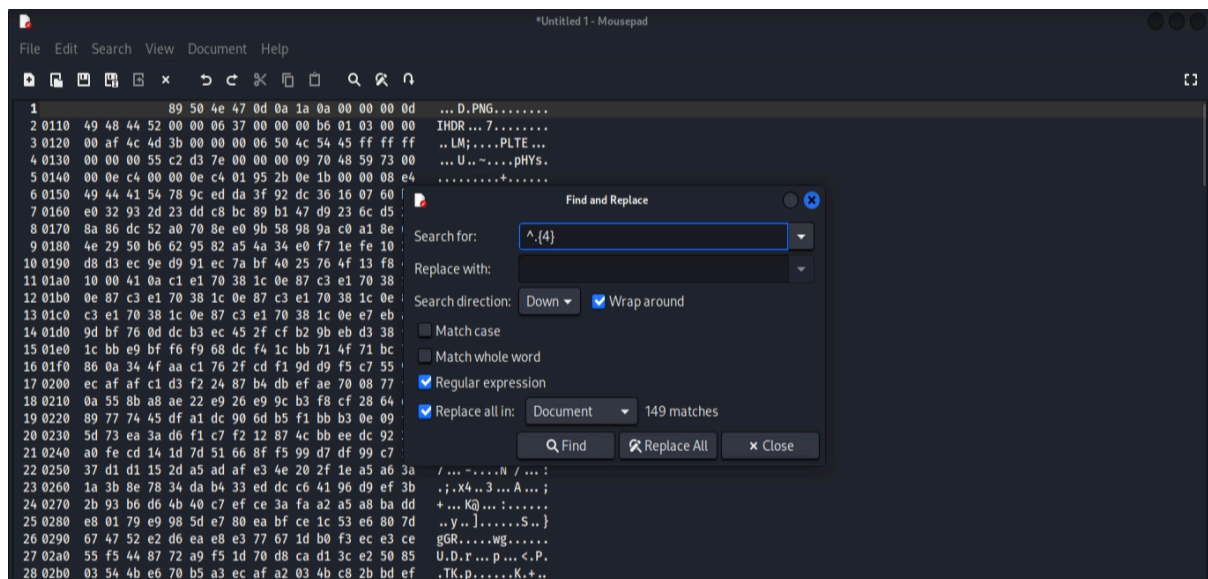


So, let's extract the file manually.

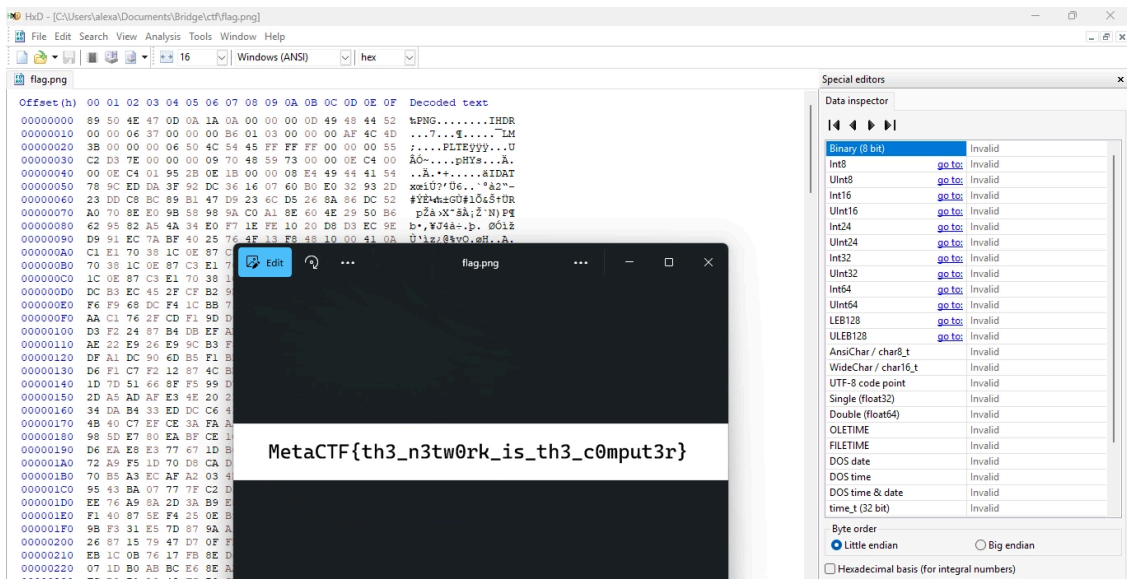
Performed a quick search in TShark by filtering only NFS traffic and using the `-x` option to display the packet data in both hexadecimal and ASCII format.



Used a regular expression to remove the packet numbers and stripped the trailing ASCII data.



Finally, copied the hexadecimal data into **HxD (Hex Editor)** and saved the reconstructed file as a PNG image and there we go.



flag: `MetaCTF{th3_n3tw0rk_is_th3_c0mput3r}`

reference:

<https://wiki.wireshark.org/RPC>

<https://www.geeksforgeeks.org/operating-systems/remote-procedure-call-rpc-in-operating-system/>

<https://aws.amazon.com/compare/the-difference-between-nfs-smb/>

<https://www.digihunch.com/2020/07/nfs-network-file-system-and-rpc-remote-procedure-call/>